Sintered NdFeB Magnet Properties

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The N30-N55 properties defined in FEMM are meant to be representative of typical examples rather than strictly based on the products from one vendor. The properties were generated using information from https://en.wikipedia.org/wiki/Neodymium_magnet. This Wikipedia page defines the following properties:

- Relative Permeability (μ_r) = 1.05
- Electrical Resistivity = 110–170 $\mu\Omega$·cm

An electrical resistivity of 150 $\mu\Omega$·cm was selected as a reasonable median value in a fairly broad range. The corresponding electrical conductivity required by FEMM is $\sigma = 0.667$ MS/m.

On the basis of data from many vendors, the numerical designation of the magnet grade represents the best-case energy product in units of MGOe as measured at room temperature (20°C) for a range of energy product usually spanning a range of 2 MGOe. However, magnets are rarely delivered magnetized to this best-case magnetization. For this reason, it is assumed that the actual energy product is the numerical grade minus one in units of MGOe, *i.e.* an N42 magnet has an energy product of 41MGOe at 20°C.

Defining BH_{max} as the energy product of the magnet and assuming a straight-line demagnetization curve, the magnet's normal coercivity, H_{cB} is defined by:

$$H_{cB} = 2\sqrt{\frac{BH_{max}}{\mu_r \mu_o}}$$

where μ_o represents the magnetic permeability of free space. Then, the remanent magnetization, B_r is defined as:

$$B_r = \mu_r \mu_o H_{cB}$$

For grades N48 and higher, there is a noticeable knee in the second-quadrant demagnetization curve. For N48, N50, N52, and N55, the location of the knee was selected on the basis of examining various manufacturer's specs. The resulting demagnetization curves are shown below in Figure 1.

The result list of NdFeB grades and representative material properties is summarized below in Table 1.

References

Arnold Magnetic Technologies, Neodynium Iron Boron Magnet Catalog, accessed 13 May 2020.
Dexter Magnetic Technologies, Dexter Catalog, accessed 23 Jan 2022.
Electron Energy Corporation Neodymium Iron Boron Magnets, accessed 10 Oct 2025.
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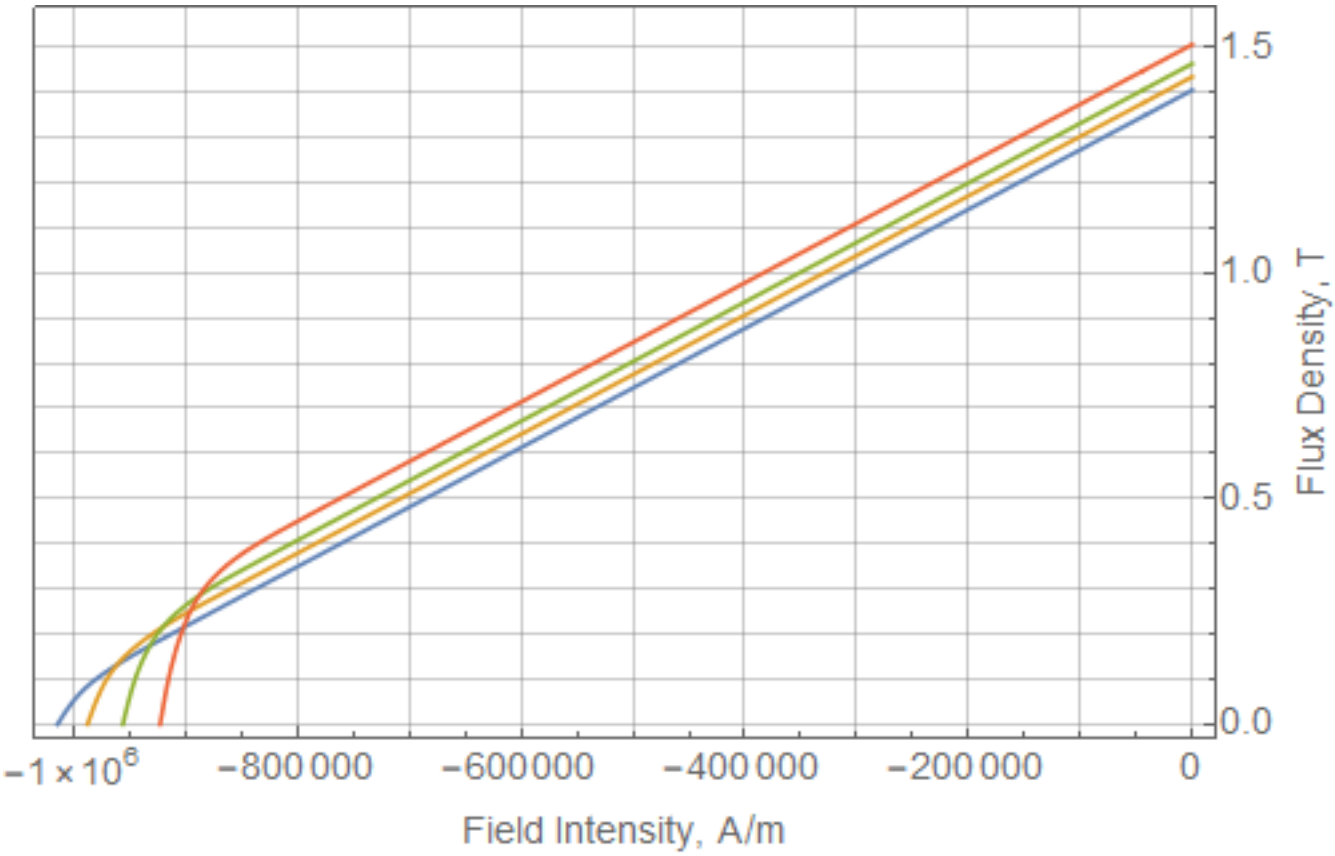


Figure 1: Assumed demagnetization curves of N48, N50, N52, and N55

Grade	BH_{max}, MGOe	B_r, T	B_r, kG	H_{cB}, kA/m	H_{cB}, kOe	σ, MS/m
N30	29	1.104	11.04	836	10.51	0.667
N33	32	1.159	11.59	879	11.04	0.667
N35	34	1.195	11.95	906	11.38	0.667
N38	37	1.247	12.47	945	11.87	0.667
N40	39	1.280	12.80	970	12.19	0.667
N42	41	1.312	13.12	995	12.50	0.667
N45	44	1.359	13.59	1030	12.95	0.667
N48	47	1.405	14.05	1015	12.75	0.667
N50	49	1.435	14.35	988	12.41	0.667
N52	51	1.464	14.64	956	12.02	0.667
N55	54	1.506	15.06	923	11.60	0.667

Table 1: Representative NdFeB Material Grades at 20°C.